

Topic model and topic codebook

```
tx <- list(
  seed = 2026L,
  min_docfreq = 5L, # DTM vocabulary floor
  run_searchK = FALSE, # TRUE for the long K-selection run
  K_grid = seq(5, 30, 3), # widen for a larger corpus
  K = 12L, # set from the searchK diagnostics
  spline_df = 5L,
  run_sensitivity = FALSE, # TRUE: refit STM at other parse floors
  sens_factors = c(0.5, 1.5), # floor multiples tested against the baseline
  facets_per_page = 9L, # topic trend panels per page
  workers = 16L,
  use_openrouter = TRUE, # FALSE to use the OpenAI-compatible endpoint
  # base_url_env and api_key_env are the NAMES of .Renviro variables, so the
  # small and large models can use different credentials and nothing
  # sensitive appears here. Ignored when use_openrouter = TRUE.
  steps = tibble::tribble(
    ~step, ~base_url_env, ~api_key_env, ~model,
    "label", "COMPASS_SMALL_URL", "COMPASS_SMALL_KEY", "MODEL-FOR-LABELING",
    "manager", "COMPASS_LARGE_URL", "COMPASS_LARGE_KEY", "MODEL-FOR-MANAGER")
  )

suppressPackageStartupMessages({
  library(here); library(stm); library(quanteda); library(tidytext)
  library(splines); library(furrr); library(viridis)
  library(tidyverse) # tidyverse LAST so its verbs win
})
# Used namespaces: clue (Hungarian matching), ellmer, glue.
theme_set(theme_minimal(base_size = 12))
options(knitr.table.max.print = Inf, knitr.kable.max_rows = Inf,
  max.print = 1e6)
set.seed(tx$seed)
`%||` <- function(x, y) if (is.null(x)) y else x
p_out <- function(f) here::here("outputs", f)
source(here::here("code", "llm_source.R"))
if (isTRUE(tx$use_openrouter)) {
  source(here::here("code", "llm_openrouter.R"))
  tx$steps$model <- open_router_models[tx$steps$step]
}
```

Input

The corpus is the paragraph units from 01_preprocess.qmd. Every cache below is fingerprinted against this file, so a changed parse cannot silently meet an old fit.

```

para_in <- p_out("paragraphs.csv")
sens_in <- p_out(c("paragraphs.csv", "blocks.csv", "linkage_plan.rds",
                  "block_embeddings.rds"))
invisible(fingerprint(para_in)) # stops by name if script 1 has not been rendered
paragraphs <- readr::read_csv(para_in, show_col_types = FALSE)
stopifnot("paragraphs.csv must carry the script 1 columns" =
          all(c("para_uid", "atom_id", "para_id", "Year", "n_tokens",
                "text") %in% names(paragraphs)))
writeLines(glue::glue("{nrow(paragraphs)} paragraph units from \\
                      {dplyr::n_distinct(paragraphs$atom_id)} documents."))

```

1484 paragraph units from 60 documents.

Tokenization

The DTM is built from cleaned tokens. Embeddings in script 1 used the raw text; this cleaning affects topics only.

```

# DTM cleaning (embeddings used the raw text; this affects topics only).
# Tokens must be purely alphabetic, which drops numbers, dates written in
# digits, and anything with special characters. Month and weekday names are
# stopped too: temporal words let topics encode Year directly, which would
# leak time into the prevalence model used to estimate trends over time.
temporal <- tibble(word = str_to_lower(c(month.name, month.abb,
    "monday", "tuesday", "wednesday", "thursday", "friday", "saturday",
    "sunday")))
tokens <- paragraphs |>
  dplyr::select(para_uid, text) |>
  tidytext::unnest_tokens(word, text) |>
  dplyr::mutate(word = str_remove_all(word, "'")) |>
  dplyr::filter(str_detect(word, "^[a-z]{2,}$")) |>
  dplyr::anti_join(tidytext::get_stopwords(), by = "word") |>
  dplyr::anti_join(temporal, by = "word")

```

Document-term matrix

Rare terms (under 5 paragraphs) are trimmed to stabilize estimation. meta is built from the surviving dfm rownames, so alignment with the model survives any dropped document. At work, check `nfeat(dfm_para)`: a vocabulary in the low thousands is comfortable at this scale, and a much larger one argues for a higher floor.

```

dfm_para <- tokens |>
  dplyr::count(para_uid, word) |>
  tidytext::cast_dfm(para_uid, word, n) |>
  quanteda::dfm_trim(min_docfreq = tx$min_docfreq)
stm_in <- quanteda::convert(dfm_para, to = "stm")
# as.data.frame, not a tibble: stm subsets metadata with base [ , ] semantics

```

```

# when building the prevalence model matrix, and a tibble returns a column
# rather than a vector there ("Error creating model matrix").
meta <- tibble(para_uid = rownames(dfm_para)) |>
  dplyr::left_join(dplyr::select(paragraphs, para_uid, atom_id, para_id, Year,
                                n_tokens, text), by = "para_uid") |>
  as.data.frame()
stopifnot("every dfm row must match a paragraph" = !anyNA(meta$Year),
  "meta must align with the stm documents" =
    nrow(meta) == length(stm_in$documents))

# Spline basis as ORDINARY COLUMNS, so no function call ever appears inside a
# formula. stm builds its prevalence model matrix by evaluating the formula
# against the metadata, and a call there (s(), ns(), or anything namespaced)
# is the documented cause of "Error creating model matrix". prev_form then
# reads ~ yr1 + yr2 + ..., which is also what estimateEffect uses below.
bs <- splines::ns(meta$Year, df = tx$spline_df)
colnames(bs) <- str_c("yr", seq_len(ncol(bs)))
meta <- cbind(meta, as.data.frame(bs))
prev_rhs <- str_c(colnames(bs), collapse = " + ") # "yr1 + yr2 + ..."
prev_form <- as.formula(str_c("~ ", prev_rhs))

```

Choosing K

stm's own parallelism forks, which is unavailable on Windows, so the held-out grid is parallelized explicitly with furrr. Read the panels together: take the smallest K where held-out likelihood has flattened and residual dispersion has stopped falling, then break ties on the coherence-exclusivity frontier. If the frontier is still improving at the top of the grid, widen tx\$K_grid rather than extrapolating. Semantic coherence is $\sum_{i < j} \log\{(D(v_i, v_j) + 1)/D(v_i)\}$ over each topic's top words, with the denominator the higher-ranked word.

```

if (isTRUE(tx$run_searchK)) {
  future::plan(future::multisession, workers = tx$workers)
  ho <- stm::make.heldout(stm_in$documents, stm_in$vocab, seed = tx$seed)
  metrics <- cache_guarded(p_out("searchK_metrics.csv"), para_in, {
    furrr::future_map(tx$K_grid, function(k) {
      fit <- stm::stm(ho$documents, ho$vocab, K = k, init.type = "Spectral",
                     prevalence = prev_form, data = meta, verbose = FALSE)
      tibble(K = k,
             heldout = stm::eval.heldout(fit, ho$missing)$expected.heldout,
             dispersion = stm::checkResiduals(fit, ho$documents)$dispersion,
             coherence = mean(stm::semanticCoherence(fit, ho$documents)),
             exclusivity = mean(stm::exclusivity(fit)))
    }, .options = furrr::furrr_options(seed = tx$seed)) |> purrr::list_rbind()
  })
  future::plan(future::sequential)
  metrics |>
    tidyr::pivot_longer(-K) |>

```

```

ggplot(aes(K, value)) + geom_line(color = viridis(1, begin = 0.35)) +
  geom_point() + facet_wrap(~ name, scales = "free_y") + labs(y = NULL)
} else {
  if (is.null(tx$K))
    stop("tx$K is NULL and tx$run_searchK is FALSE. Re-run the config chunk, ",
         "or set tx$K before continuing.")
  glue::glue("searchK skipped (tx$run_searchK = FALSE); K fixed at {tx$K}")
}

```

```
searchK skipped (tx$run_searchK = FALSE); K fixed at 12
```

Structural topic model

Year enters the prevalence model as a spline (df 5), so topic attention is allowed to drift over the period instead of being averaged away; spectral initialization is deterministic given the data, removing initialization luck from a fit everything downstream depends on.

```

stm_fit <- cache_guarded(p_out("stm_fit.rds"), para_in,
  stm::stm(stm_in$documents, stm_in$vocab, K = tx$K, init.type = "Spectral",
    prevalence = prev_form, data = meta, verbose = FALSE))

# K comes from the FITTED model, not the config: a cached fit can predate a
# change to tx$K, and every downstream object is sized by the fit.
K_fit <- ncol(stm_fit$theta)
if (K_fit != tx$K)
  stop("Cached stm_fit.rds has K = ", K_fit, " but tx$K = ", tx$K,
       ". Delete outputs/stm_fit.rds (and stm_year_effects.rds, ",
       "topic_trends.csv) once, then re-render.")

theta <- stm_fit$theta
colnames(theta) <- str_c("topic_", seq_len(K_fit))
paragraph_theta <- meta |> dplyr::bind_cols(tibble::as_tibble(theta))

```

Modal topic assignment and its diagnostics

Each paragraph is assigned its **modal** topic, so topics partition the corpus (mutually exclusive, collectively exhaustive) and can serve as sampling strata in Script 2. This is the same move as modal class assignment in latent class analysis, and it inherits the same caveat: a modal assignment is an estimate, not an observation, and treating it as observed pushes classification error downstream. Two LCA diagnostics report how much error to expect. **Entropy** R^2 summarizes overall assignment sharpness on a 0-1 scale; **AvePP**, the average posterior probability among a topic's own assigned paragraphs, does it per topic, with about 0.7 the conventional floor. Unlike LCA, STM supplies no classification error matrix over modal assignment, so no BCH-style correction is available here; the honest substitutes are the min_theta scope floor in Script 2 and its sensitivity analysis.

```

paragraph_theta <- paragraph_theta |>
  dplyr::mutate(topic = max.col(theta, ties.method = "first"),
               theta_max = theta[cbind(dplyr::row_number(), topic)])

ent <- -sum(theta * log(theta + 1e-12))
entropy_r2 <- 1 - ent / (nrow(theta) * log(K_fit))

topic_summary <- tibble(
  topic = seq_len(K_fit),
  prevalence = colMeans(theta),
  frex = apply(stm::labelTopics(stm_fit, n = 8)$frex, 1, str_c,
               collapse = ", "),
  prob = apply(stm::labelTopics(stm_fit, n = 8)$prob, 1, str_c,
               collapse = ", ")) |>
  dplyr::left_join(
    paragraph_theta |>
      dplyr::group_by(topic) |>
      dplyr::summarise(n paras = dplyr::n(), avepp = mean(theta_max)),
    by = "topic")

readr::write_csv(topic_summary, p_out("topic_summary.csv"))
readr::write_csv(paragraph_theta, p_out("paragraph_theta.csv"))
theta_in <- p_out("paragraph_theta.csv")

glue::glue("Entropy R2 = {round(entropy_r2, 3)}; topics with AvePP < 0.7: ",
           "{sum(topic_summary$avepp < 0.7)} of {K_fit}")

```

Entropy R2 = 0.471; topics with AvePP < 0.7: 12 of 12

```

topic_summary |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  dplyr::select(topic, prevalence, n paras, avepp) |>
  tbl_paged(caption = "Topics by prevalence, with assigned paragraph counts and
AvePP (below 0.7 means fuzzy modal assignment).")

```

Topics by prevalence, with assigned paragraph counts and AvePP (below 0.7 means fuzzy modal assignment).

topic	prevalence	n paras	avepp
6	0.148	269	0.523
2	0.106	136	0.552
5	0.102	154	0.574
3	0.095	117	0.504
9	0.089	122	0.544

topic	prevalence	n_paras	avepp
11	0.082	126	0.617
1	0.077	104	0.510
7	0.071	99	0.572
4	0.065	111	0.540
10	0.060	99	0.613
8	0.053	74	0.547
12	0.052	73	0.609

```
topic_summary |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  dplyr::mutate(ttl = str_c("Topic ", topic)) |>
  tbl_records("ttl", c("frex", "prob"),
    caption = "Topic terms: the evidence for checking each label.")
```

Topic terms: the evidence for checking each label.

Topic 6 - frex: let, americans, americas, together, god, heart, story, bless - prob: us, let, america, can, must, one, americans, work

Topic 2 - frex: thank, day, promise, years, voices, crisis, four, many - prob: nation, years, day, many, people, great, now, government

Topic 5 - frex: peace, international, nations, peoples, practice, arms, effort, settlement - prob: peace, world, nations, can, free, new, shall, freedom

Topic 3 - frex: opportunity, problems, ideals, american, dignity, society, liberty, equality - prob: american, people, government, must, liberty, freedom, nation, every

Topic 9 - frex: vice, health, change, across, live, mr, president, weapons - prob: new, president, change, world, life, citizens, live, nation

Topic 11 - frex: discharge, presence, office, countrymen, duties, oath, meaning, station - prob: shall, citizens, fellow, duties, people, upon, office, country

Topic 1 - frex: purpose, rules, pride, helped, face, task, save, true - prob: can, national, men, purpose, upon, spirit, good, great

Topic 7 - frex: federal, authority, constitution, granted, powers, legislative, measures, terms - prob: government, constitution, states, power, people, union, shall, one

Topic 4 - frex: amendment, laws, law, party, negro, enforce, enforced, combinations - prob: law, laws, government, party, public, people, service, can

Topic 10 - frex: intercourse, extended, harmony, limits, union, happy, cultivate, agitation - prob: government, union, states, every, people, country, spirit, great

Topic 8 - frex: enforcement, navy, naval, army, military, coast, invasion, civil - prob: war, states, foreign, united, force, country, us, great

Topic 12 - frex: tariff, revenue, currency, expenditures, debt, money, taxation, credit - prob: public, government, congress, revenue, upon, country, business, people

Does the parse drive the topics?

The floor is calibrated to the corpus, but the floor is still a decision about the document unit, and the topic model is sensitive to document length. This section tests whether that choice matters. The unparagraphed documents are relinked at 0.5 and 1.5 times the floor calibrated in script 1, using the vectors cached there, so no API calls are made, the STM is refit at the same K on each, and every baseline topic is matched one-to-one to its best counterpart in each alternative fit by the cosine of their topic-word distributions over the shared vocabulary (Hungarian assignment via `clue::solve_LSAP`).

The result is reported as similarities, not as a pass or fail, since any threshold would be invented. Read it this way: if most matched cosines are high at both alternative floors, topics are properties of the corpus and not of the parse; if a few topics match poorly, those are the ones whose existence depends on how text was grouped, and they deserve caution before being carried into stance measurement.

```
if (isTRUE(tx$run_sensitivity)) {
  blocks <- readr::read_csv(p_out("blocks.csv"), show_col_types = FALSE)
  plan <- readRDS(p_out("linkage_plan.rds"))
  emb <- readRDS(p_out("block_embeddings.rds"))
  if (!any(!plan$doc$paragraphed))
    writeLines("No documents were relinked in script 1; the test has nothing to
vary.")
  refit <- function(f) {
    p <- link_corpus(blocks, emb$vec, plan$floor * f, plan$doc)
    d <- p |>
      dplyr::select(para_uid, text) |>
      tidytext::unnest_tokens(word, text) |>
      dplyr::mutate(word = str_remove_all(word, "'")) |>
      dplyr::filter(str_detect(word, "^[a-z]{2,}$")) |>
      dplyr::anti_join(tidytext::get_stopwords(), by = "word") |>
      dplyr::anti_join(temporal, by = "word") |>
      dplyr::count(para_uid, word) |>
      tidytext::cast_dfm(para_uid, word, n) |>
      quantda::dfm_trim(min_docfreq = tx$min_docfreq)
    s_in <- quantda::convert(d, to = "stm")
    list(n_docs = length(s_in$documents),
         fit = stm::stm(s_in$documents, s_in$vocab, K = K_fit,
                        init.type = "Spectral", verbose = FALSE))
  }
  sens <- cache_guarded(p_out("sensitivity_fits.rds"), sens_in,
```

```

      purrr::map(tx$sens_factors, refit) |>
      purrr::set_names(tx$sens_factors))
b_ref <- beta_matrix(stm_fit)
sens_tbl <- purrr::imap(sens, function(x, f)
  match_topics(b_ref, beta_matrix(x$fit)) |>
  dplyr::mutate(floor_factor = as.numeric(f), n_docs = x$n_docs)) |>
  purrr::list_rbind()
readr::write_csv(sens_tbl, p_out("parse_sensitivity.csv"))

sens_tbl |>
  dplyr::group_by(floor_factor) |>
  dplyr::summarise(documents = dplyr::first(n_docs),
    median_cosine = stats::median(cosine),
    lowest_cosine = min(cosine),
    shared_vocab = dplyr::first(shared_vocab),
    .groups = "drop") |>
  tbl_paged(caption = "Topic stability across parse floors: matched topic-word
cosine against the baseline fit.")
sens_tbl |>
  tidyr::pivot_wider(id_cols = topic, names_from = floor_factor,
    values_from = cosine, names_prefix = "floor_x") |>
  dplyr::left_join(dplyr::select(topic_summary, topic, prevalence),
    by = "topic") |>
  dplyr::arrange(dplyr::desc(prevalence)) |>
  tbl_paged(caption = "Per-topic matched cosine by floor multiple. Low values
mark topics whose existence depends on the parse.")
} else {
  writeLines("Sensitivity test skipped (tx$run_sensitivity = FALSE).")
}

```

Sensitivity test skipped (tx\$run_sensitivity = FALSE).

Topic codebook

The codebook is written **once**, after the final STM, and is the naming record from here on. If `topic_codebook.csv` already exists on disk, this section reads it and makes no API calls: the file is assumed to carry your edits, and re-drafting would overwrite them. Delete the file to redraft from scratch.

Two passes, deliberately shaped:

Pass 1, one call per topic with the smaller model (label step). Each call sees only that topic's own evidence: its FREX terms, its highest-probability terms, and its three highest-theta paragraphs. It is blind to every other topic, to Year, and to stance. That blindness is the point: a model shown all topics at once starts fitting a narrative across them, and labels stop being traceable to the evidence a human can check them against.

Pass 2, one manager call with the larger model (manager step). Blind drafting has one predictable failure: two topics can come back with near-duplicate labels even though their term distributions differ. The manager sees all K labels with their FREX terms and resolves collisions. Its mandate is deliberately narrow, stated in the prompt and enforced by the schema: it may **rename only**. It cannot merge topics, split them, change a description, or reorder anything, because the topic structure is the statistical model's output and an LLM should not be permitted to override it.

The proposition column is written empty here. Script 2 fills it for the one topic it processes, since propositions are stance targets and you only need them for the topics you actually analyze.

```
cb_path <- p_out("topic_codebook.csv")

# A refit can renumber topics, and a codebook written for the old fit would
# then label the wrong topics silently. The codebook's fingerprint is taken on
# the theta it was drafted from; your edits to the CSV itself never trip it.
check_fingerprint(cb_path, theta_in, strict = FALSE)
if (file.exists(cb_path)) {
  codebook <- readr::read_csv(cb_path, show_col_types = FALSE)
  writelines(glue::glue("Existing codebook found ({nrow(codebook)} topics); \\
                        no LLM calls made. Delete {basename(cb_path)} to \\
                        redraft."))
} else {
  ev <- topic_summary |>
  dplyr::left_join(
    paragraph_theta |>
    dplyr::group_by(topic) |>
    dplyr::slice_max(theta_max, n = 3) |>
    dplyr::summarise(ex = str_c(str_trunc(text, 300), collapse = "\n- "),
                    .groups = "drop"),
    by = "topic")

# ---- Pass 1: one blind call per topic -----
label_sys <- build_prompt(
  role = str_c("You are a survey methodologist who names statistical topics",
    ",
    "so colleagues can use them."),
  task = str_c("Read this topic's distinctive terms and example paragraphs.",
    ",
    "Write a short label and one factual sentence describing what",
    ",
    "these paragraphs are about. Before answering, check that ",
    "every word of your description is supported by the evidence ",
    "shown."),
  rules = c("Use only the terms and paragraphs shown.",
    "Do not infer stance, sentiment, intent, or who is speaking.",
    "Do not invent statistics, dates, places, or named entities.",
    "If the evidence is mixed, describe the mixture plainly."),
  output = "A label of at most 4 words and a one-sentence description.")
```

```

label_schema <- ellmer::type_object(
  label      = ellmer::type_string("Topic name, at most 4 words."),
  description = ellmer::type_string("One factual sentence.")
)
label_chat <- make_chat("label", label_sys, tx$steps, tx$use_openrouter)

drafts <- purrr::map(seq_len(nrow(ev)), function(i) {
  r <- call_one(label_chat,
    str_c("Distinctive (FREX) terms: ", ev$frex[i],
      "\nHighest-probability terms: ", ev$prob[i],
      "\nExample paragraphs:\n- ", ev$ex[i]), label_schema)
  tibble(topic = ev$topic[i],
    label = r$label %||% NA_character_,
    description = r$description %||% NA_character_)
}) |> purrr::list_rbind()
if (anyNA(drafts$label))
  warning(sum(is.na(drafts$label)), " topic label calls failed; those rows ",
    "are blank in the codebook and need a name by hand.")

# ---- Pass 2: manager resolves duplicate labels, renaming ONLY -----
mgr_sys <- build_prompt(
  role = str_c("You are a senior methodologist reviewing a topic codebook ",
    "for clarity before it goes to coders."),
  task = str_c("You are shown every topic's draft label with its ",
    "distinctive terms. Some labels are near-duplicates of each ",
    "other even though the topics differ. Return a label for ",
    "EVERY topic: unchanged where it is already distinct, and ",
    "revised only where two or more labels would be confused. ",
    "Before answering, check that no two returned labels are ",
    "the same and that every topic number appears exactly once."),
  rules = c("Rename only. Never merge, split, drop, or reorder topics.",
    "Return the same number of topics you were given.",
    "Base any rename on the distinctive terms shown, nothing else.",
    "Keep labels to at most 4 words.",
    "Do not write descriptions; they are not yours to change."),
  output = "One label per topic number.")
mgr_schema <- ellmer::type_object(
  topics = ellmer::type_array(
    items = ellmer::type_object(
      topic = ellmer::type_integer("The topic number, unchanged."),
      label = ellmer::type_string("Final label, at most 4 words.)))
mgr_prompt <- str_c(
  "Draft labels with their distinctive terms:\n",
  str_c("- topic ", drafts$topic, ": \"", drafts$label, "\" | terms: ",
    ev$frex[match(drafts$topic, ev$topic)], collapse = "\n"))
mgr <- call_one(make_chat("manager", mgr_sys, tx$steps, tx$use_openrouter),
  mgr_prompt, mgr_schema)

# The manager is advisory: its output is applied ONLY if it returns exactly

```

```

# one label per topic with no topics invented or lost. Any deviation and the
# blind drafts stand, because a malformed rename must never reshape the
# codebook silently.
final <- if (!is.null(mgr) && length(mgr$topics) == nrow(drafts)) {
  renamed <- purrr::map(mgr$topics, ~ tibble(topic = as.integer(.x$topic),
                                             label_new = .x$label)) |>

  purrr::list_rbind()
  if (setequal(renamed$topic, drafts$topic)) {
    drafts |> dplyr::left_join(renamed, by = "topic") |>
      dplyr::mutate(label = dplyr::coalesce(label_new, label)) |>
      dplyr::select(-label_new)
  } else drafts
} else drafts
n_renamed <- sum(final$label != drafts$label, na.rm = TRUE)

codebook <- final |>
  dplyr::mutate(proposition = NA_character_) |> # Script 2 fills this
  dplyr::left_join(topic_summary |> dplyr::select(topic, prevalence, avepp),
                  by = "topic") |>
  dplyr::arrange(topic)
readr::write_csv(codebook, cb_path)
write_fingerprint(cb_path, theta_in)
writeLines(glue::glue("Codebook drafted: {nrow(codebook)} topics, \\
                      {n_renamed} renamed by the manager. EDIT \\
                      {basename(cb_path)} before running Script 2; it will \\
                      not be redrafted while the file exists."))
}

```

Codebook drafted: 12 topics, 0 renamed by the manager. EDIT topic_codebook.csv before running Script 2; it will not be redrafted while the file exists.

```

codebook |>
  dplyr::mutate(ttl = str_c("Topic ", topic, ": ", label)) |>
  tbl_records("ttl", "description",
             caption = "Topic codebook. Labels drafted blind per topic, then
deduplicated by a rename-only manager pass. Edit the CSV; nothing downstream
redrafts it.")

```

Topic codebook. Labels drafted blind per topic, then deduplicated by a rename-only manager pass. Edit the CSV; nothing downstream redrafts it.

Topic 1: National Restoration - description: These paragraphs are about a national effort to restore and rebuild after a period of failure and decline.

Topic 2: Inaugural Address - description: These paragraphs are about a government leader's speech on a significant day, referencing the nation's past and promising progress amidst crisis.

Topic 3: American Society - description: These paragraphs discuss American values and government in relation to liberty, freedom, and the nation's ability to solve problems and protect its people.

Topic 4: Law Enforcement - description: The paragraphs discuss the enforcement of laws, including amendments, and their impact on the negro race and the country.

Topic 5: International Peace - description: These paragraphs are about efforts to achieve peace and prosperity in the world through international cooperation and the application of technical knowledge.

Topic 6: American Renewal - description: These paragraphs are about Americans working together to address challenges and renew their country.

Topic 7: Constitutional Powers - description: These paragraphs are about the distribution of powers between the federal government and the states as granted by the constitution.

Topic 8: Military Defense - description: These paragraphs are about the importance of maintaining a strong military force, including the navy and army, to secure the country against invasion and other dangers.

Topic 9: Presidential Speech - description: These paragraphs are about presidential speeches that mention offers of unity and change to fellow citizens and other government leaders, including a vice president and a speaker.

Topic 10: Government Union - description: The paragraphs discuss the importance of harmony and balance within the government and union of a country.

Topic 11: Inaugural Address - description: These paragraphs are about a public official expressing gratitude and acknowledging the trust placed in them by citizens upon assuming a high office.

Topic 12: Government Finance - description: These paragraphs are about government revenue, taxation, and expenditures.

Representative paragraphs

```
top4 <- topic_summary |> dplyr::slice_max(prevalence, n = 4) |>
  dplyr::pull(topic)
th <- stm::findThoughts(stm_fit, texts = meta$para_uid, n = 3, topics = top4)
tibble(topic = rep(top4, each = 3), para_uid = unlist(th$docs)) |>
  dplyr::left_join(paragraphs |> dplyr::select(para_uid, text),
    by = "para_uid") |>
  dplyr::mutate(ttl = str_c("Topic ", topic, " | ", para_uid)) |>
  tbl_records("ttl", "text",
    caption = "Three highest-theta paragraphs for the four most
prevalent topics.")
```

Three highest-theta paragraphs for the four most prevalent topics.

Topic 6 | 1993-Clinton-15 - text: My fellow Americans, you, too, must play your part in our renewal. I challenge a new generation of YOUNG Americans to a season of service, to act on your idealism, by helping troubled children, keeping company with those in need, reconnecting our torn communities. There is so much to be done. Enough, indeed, for millions of others who are still young in spirit, to give of themselves in service, too. In serving we recognize a simple, but powerful, truth: we need each other, and we must care for one another. Today we do more than celebrate America, we rededicate ourselves to the very idea of America, an idea born in revolution, and renewed through two centuries of challenge, an idea tempered by the knowledge that but for fate, we, the fortunate and the unfortunate, might have been each other; an idea ennobled by the faith that our nation can summon from its myriad diversity, the deepest measure of unity; an idea infused with the conviction that America's journey long, heroic journey must go forever upward.

Topic 6 | 2009-Obama-22 - text: Recall that earlier generations faced down fascism and communism not just with missiles and tanks, but with the sturdy alliances and enduring convictions. They understood that our power alone cannot protect us, nor does it entitle us to do as we please. Instead, they knew that our power grows through its prudent use; our security emanates from the justness of our cause, the force of our example, the tempering qualities of humility and restraint.

Topic 6 | 2009-Obama-35 - text: America! In the face of our common dangers, in this winter of our hardship, let us remember these timeless words. With hope and virtue, let us brave once more the icy currents, and endure what storms may come. Let it be said by our children's children that when we were tested we refused to let this journey end, that we did not turn back nor did we falter; and with eyes fixed on the horizon and God's grace upon us, we carried forth that great gift of freedom and delivered it safely to future generations.

Topic 2 | 1989-Bush-03 - text: I have just repeated word for word the oath taken by George Washington 200 years ago, and the Bible on which I placed my hand is the Bible on which he placed his. It is right that the memory of Washington be with us today, not only because this is our Bicentennial Inauguration, but because Washington remains the Father of our Country. And he would, I think, be gladdened by this day; for today is the concrete expression of a stunning fact: our continuity these 200 years since our government began.

Topic 2 | 1981-Reagan-22 - text: In the days ahead I will propose removing the roadblocks that have slowed our economy and reduced productivity. Steps will be taken aimed at restoring the balance between the various levels of government. Progress may be slow - measured in inches and feet, not miles - but we will progress. Is it time to reawaken this industrial giant, to get government back within its means, and to lighten our punitive tax burden. And these will be our first priorities, and on these principles, there will be no compromise.

Topic 2 | 1969-Nixon-10 - text: Our crisis today is the reverse. We have found ourselves rich in goods, but ragged in spirit; reaching with magnificent precision for the moon, but falling into raucous discord on earth. We are caught in war, wanting peace. We are torn by division, wanting unity. We see around us empty lives, wanting fulfillment. We see tasks that need doing, waiting for hands to do them. To a crisis of the spirit, we need an answer of the spirit. To find that answer,

we need only look within ourselves. When we listen to “the better angels of our nature,” we find that they celebrate the simple things, the basic things - such as goodness, decency, love, kindness. Greatness comes in simple trappings. The simple things are the ones most needed today if we are to surmount what divides us, and cement what unites us. To lower our voices would be a simple thing.

Topic 5 | 1949-Truman-13 - text: Since the end of hostilities, the United States has invested its substance and its energy in a great constructive effort to restore peace, stability, and freedom to the world. We have sought no territory and we have imposed our will on none. We have asked for no privileges we would not extend to others. We have constantly and vigorously supported the United Nations and related agencies as a means of applying democratic principles to international relations. We have consistently advocated and relied upon peaceful settlement of disputes among nations. We have made every effort to secure agreement on effective international control of our most powerful weapon, and we have worked steadily for the limitation and control of all armaments. We have encouraged, by precept and example, the expansion of world trade on a sound and fair basis.

Topic 5 | 1949-Truman-27 - text: Greater production is the key to prosperity and peace. And the key to greater production is a wider and more vigorous application of modern scientific and technical knowledge. Only by helping the least fortunate of its members to help themselves can the human family achieve the decent, satisfying life that is the right of all people. Democracy alone can supply the vitalizing force to stir the peoples of the world into triumphant action, not only against their human oppressors, but also against their ancient enemies - hunger, misery, and despair. On the basis of these four major courses of action we hope to help create the conditions that will lead eventually to personal freedom and happiness for all mankind. If we are to be successful in carrying out these policies, it is clear that we must have continued prosperity in this country and we must keep ourselves strong. Slowly but surely we are weaving a world fabric of international security and growing prosperity. We are aided by all who wish to live in freedom from fear - even by those who live today in fear under their own governments. We are aided by all who want relief from the lies of propaganda - who desire truth and sincerity. We are aided by all who desire self-government and a voice in deciding their own affairs. We are aided by all who long for economic security - for the security and abundance that men in free societies can enjoy. We are aided by all who desire freedom of speech, freedom of religion, and freedom to live their own lives for useful ends. Our allies are the millions who hunger and thirst after righteousness.

Topic 5 | 1949-Truman-23 - text: I believe that we should make available to peace-loving peoples the benefits of our store of technical knowledge in order to help them realize their aspirations for a better life. And, in cooperation with other nations, we should foster capital investment in areas needing development. Our aim should be to help the free peoples of the world, through their own efforts, to produce more food, more clothing, more materials for housing, and more mechanical power to lighten their burdens.

Topic 3 | 1921-Harding-02 - text: When one surveys the world about him after the great storm, noting the marks of destruction and yet rejoicing in the ruggedness of the things which withstood it, if he is an American he breathes the clarified atmosphere with a strange mingling of regret

and new hope. We have seen a world passion spend its fury, but we contemplate our Republic unshaken, and hold our civilization secure. Liberty - liberty within the law - and civilization are inseparable, and though both were threatened we find them now secure; and there comes to Americans the profound assurance that our representative government is the highest expression and surest guaranty of both.

Topic 3 | 1997-Clinton-06 - text: Now, for the third time, a new century is upon us, and another time to choose. We began the 19th century with a choice, to spread our nation from coast to coast. We began the 20th century with a choice, to harness the Industrial Revolution to our values of free enterprise, conservation, and human decency. Those choices made all the difference. At the dawn of the 21st century a free people must now choose to shape the forces of the Information Age and the global society, to unleash the limitless potential of all our people, and, yes, to form a more perfect union.

Topic 3 | 1937-Roosevelt-03 - text: We of the Republic sensed the truth that democratic government has innate capacity to protect its people against disasters once considered inevitable, to solve problems once considered unsolvable. We would not admit that we could not find a way to master economic epidemics just as, after centuries of fatalistic suffering, we had found a way to master epidemics of disease. We refused to leave the problems of our common welfare to be solved by the winds of chance and the hurricanes of disaster.

Prevalence over time

```
stm_year_effects <- cache_guarded(p_out("stm_year_effects.rds"), para_in,
  stm::estimateEffect(
    as.formula(str_c("1:", K_fit, " ~ ", prev_rhs)),
    stm_fit, metadata = meta, uncertainty = "Global"))
```

Expected topic prevalence is predicted across the observed year range from the fitted effect object, with 95 percent intervals. Two details make the prediction correct rather than merely plausible. The spline basis is rebuilt at the prediction grid with `predict(bs, ...)`, which reuses the **original knots**; calling `ns()` afresh on the grid would place knots differently and mis-predict silently. And the intervals come from the simulation draws `estimateEffect` already stores under `uncertainty = "Global"`, so they carry the topic-model uncertainty rather than regression standard errors alone.

```
# Prediction grid: every observed year, with the basis evaluated at the
# ORIGINAL knots so the fitted coefficients apply.
grid <- tibble(Year = sort(unique(meta$Year)))
Xg    <- cbind(1, predict(bs, newx = grid$Year))

# estimateEffect stores per-topic simulation draws of the coefficient vector
# in $parameters[[k]][[j]]$est. Predicting each draw and taking quantiles
# propagates both the regression and the topic-model uncertainty.
draws_for_topic <- function(k) {
```

```

purrr::map(stm_year_effects$parameters[[k]], function(sim)
  as.numeric(Xg %*% sim$est)) |>
  purrr::reduce(rbind) |> matrix(ncol = nrow(grid))
}
topic_trends <- purrr::map(seq_along(stm_year_effects$topics), function(i) {
  d <- draws_for_topic(i)
  grid |> dplyr::mutate(
    topic = stm_year_effects$topics[i],
    fit   = apply(d, 2, mean),
    lo    = apply(d, 2, quantile, 0.025),
    hi    = apply(d, 2, quantile, 0.975))
}) |> purrr::list_rbind()
readr::write_csv(topic_trends, p_out("topic_trends.csv"))
glue::glue("Predicted prevalence for {dplyr::n_distinct(topic_trends$topic)} \
\
      topics across {nrow(grid)} years written to topic_trends.csv")

```

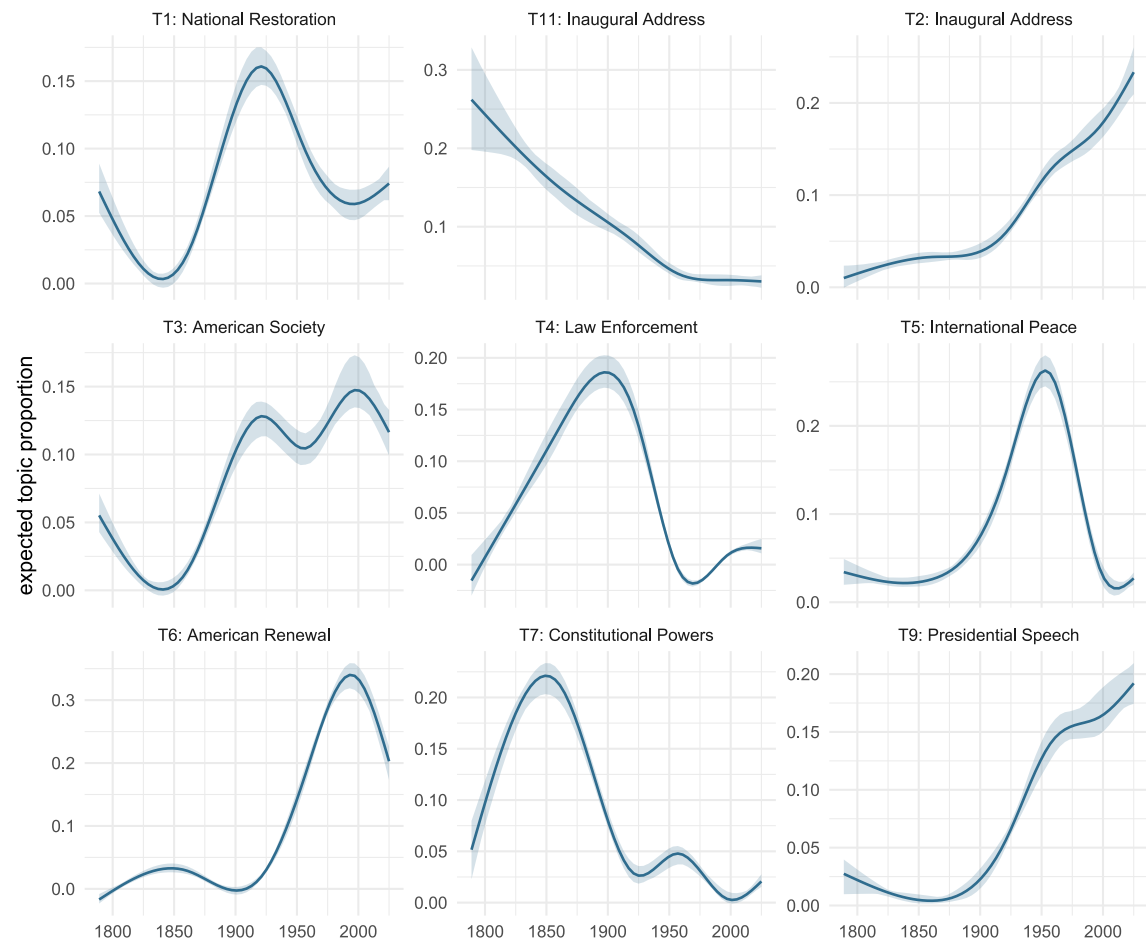
Predicted prevalence for 12 topics across 60 years written to topic_trends.csv

```

# All topics, paged so panels stay legible: 25 facets on one page renders each
# about 1.5 inches wide and the ribbons become unreadable.
trend_panel <- function(d, page, n_pages) {
  ggplot(d, aes(Year, fit)) +
    geom_ribbon(aes(ymin = lo, ymax = hi), alpha = 0.2,
              fill = viridis(1, begin = 0.35)) +
    geom_line(linewidth = 0.7, color = viridis(1, begin = 0.35)) +
    facet_wrap(~ topic_label, scales = "free_y", ncol = 3) +
    labs(x = NULL, y = "expected topic proportion",
         title = glue::glue("Topic prevalence over time ({page} of {n_pages})"),
         subtitle = "Model-based estimates with 95 percent intervals")
}
trend_pages <- topic_trends |>
  dplyr::left_join(topic_summary |> dplyr::select(topic, prevalence),
                  by = "topic") |>
  dplyr::left_join(codebook |> dplyr::select(topic, label), by = "topic") |>
  dplyr::mutate(
    topic_label = str_c("T", topic, ":", str_trunc(label, 22)),
    page = (dplyr::dense_rank(dplyr::desc(prevalence)) - 1L) %/%
           tx$facets_per_page + 1L) |>
  dplyr::group_split(page)
purrr::iwalk(trend_pages, function(d, i)
  print(trend_panel(d, i, length(trend_pages))))

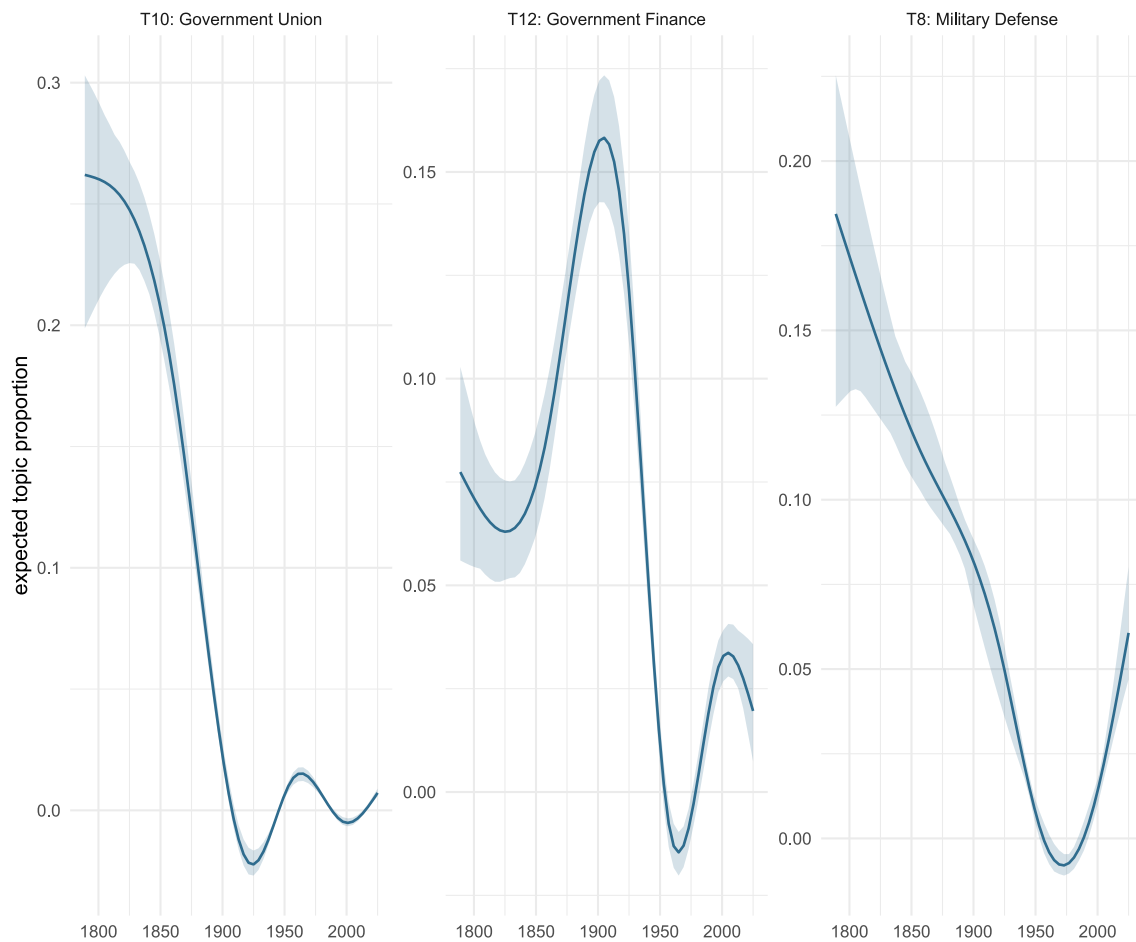
```

Topic prevalence over time (1 of 2)
Model-based estimates with 95 percent intervals



Topic prevalence over time (2 of 2)

Model-based estimates with 95 percent intervals



```
glue::glue("outputs/ written: paragraphs.csv, paragraph_theta.csv (ids, Year, \
\
n_tokens, text, topic_1..K, topic, theta_max), topic_summary.csv, \
topic_codebook.csv (labels; 03_stance.qmd adds propositions), topic_trends.csv, \
stm_fit.rds, stm_year_effects.rds. 03_stance.qmd reads these.")
```

```
outputs/ written: paragraphs.csv, paragraph_theta.csv (ids, Year, n_tokens,
text, topic_1..K, topic, theta_max), topic_summary.csv, topic_codebook.csv
(labels; 03_stance.qmd adds propositions), topic_trends.csv,
stm_fit.rds, stm_year_effects.rds. 03_stance.qmd reads these.
```